

Table of Contents

Deliverable 2	Error! Bookmark not defined.
Criteria.....	Error! Bookmark not defined.
Introduction	3
Prototype	Error! Bookmark not defined.
Main Website	4
Login.....	4
Sign up.....	5
Main Pages.....	6
Your Cart, Checkout and Your Orders	8
Subscription and edit information	10
Product Page.....	11
Admin.....	Error! Bookmark not defined.
Goals & Objectives	13
Class Responsibility Collaborator (CRC) Cards	13
Enhanced Entity Relationship Diagram (EERD)	14
Context Diagram	15
Data Flow Diagram (DFD)	16
Use Case Diagram	17
Coding	18
Screenshot	18
PHP	19
HTML.....	20
CSS	21
JavaScript.....	22
MySQL Sample Tables.....	24
Conclusion	Error! Bookmark not defined.
Table of Figures	26
Bibliography	Error! Bookmark not defined.

Mission

Design a prototype for the e-commerce platform you intend to build as specified in Deliverable 1 (Project Proposal). The prototype platform will be used to by customers to purchase goods from DragonStone. Your platform must also include a website used by administrators to manage the platform. The admin website should cater for all the various users within the organisation. You should ensure that there are proper restrictions in place for the different roles within the organisation, and that different types of users can be created, displayed, updated, and deleted.

Provide the following diagrams as part of your documentation:

- Class Responsibility Collaborator (CRC) Cards
- The Enhanced Entity Relationship Diagram (EERD)
- Context Diagram
- Data Flow Diagram (DFD)
- Use Case Diagram

Develop the database-driven e-commerce platform that you designed using the designs mentioned above. Your database must be designed in such a way that allows for reports to be generated based on aggregated data, and your admin site should allow various users to generate reports based on their different roles within the organisation. You must include sufficient dummy data to demonstrate the full functionality of the system.

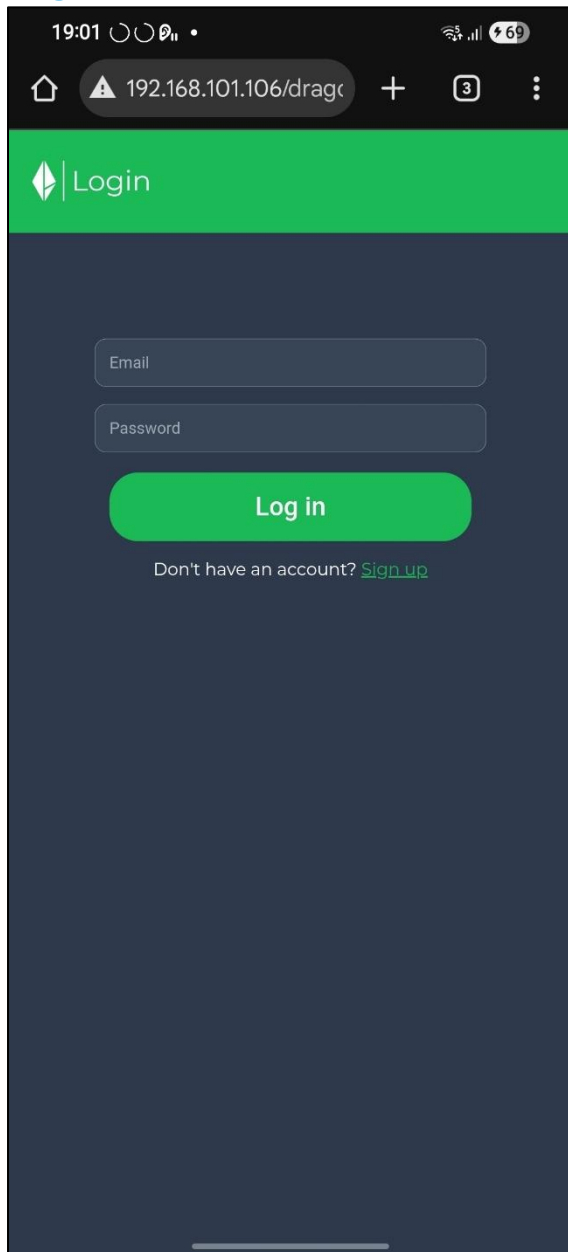
Introduction

This document transitions the conceptual framework of the DragonStone e-commerce platform into a demonstrable, database-driven prototype. Deliverable 2 establishes the foundational technical specifications and design artifacts, including the Enhanced Entity Relationship Diagram (EERD) and Data Flow Diagrams (DFD). The core objective is to validate the architectural feasibility of DragonStone's custom requirements, such as the integrated carbon footprint calculation and the EcoPoints reward system. The accompanying functional web application serves to demonstrate the essential features and the scalable, community-focused platform with administrative tools for management of the dragonstone platform.

Website Screenshots

Main Website

Login



Sign up

19:01 69

192.168.101.106/dragc

Sign up

First name

Last name

Email

Cell number

Password 

Password Requirements:

- Minimum 8 characters
- At least 1 uppercase letter (A-Z)
- At least 1 number (0-9)
- At least 1 symbol (!@#\$%^&*)

Already have an account? [Log in](#)

Next: Address

19:03 71

192.168.101.106/dragc

Sign up

Street number & name

Suburb

City

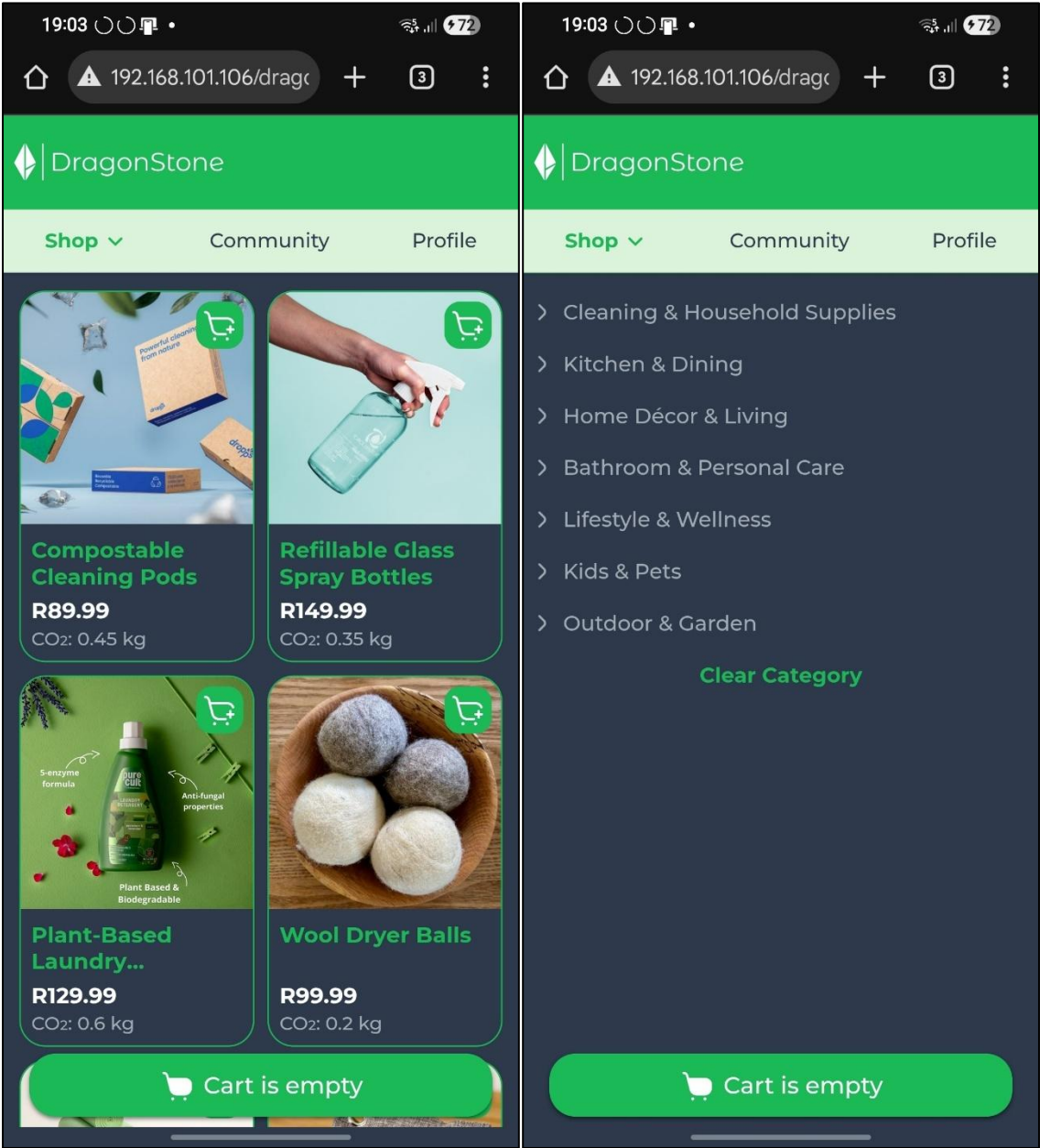
Province

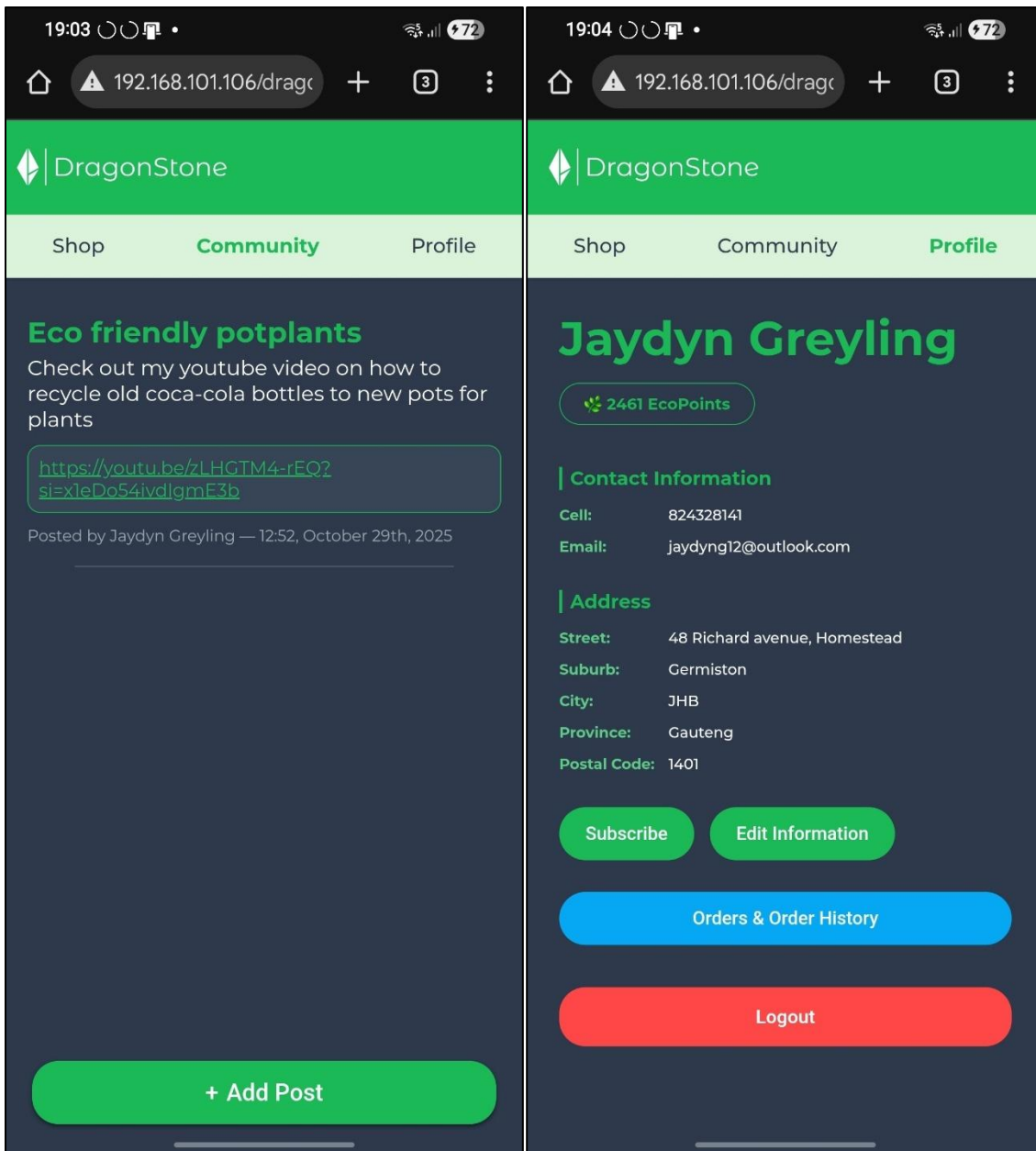
Postal Code

Already have an account? [Log in](#)

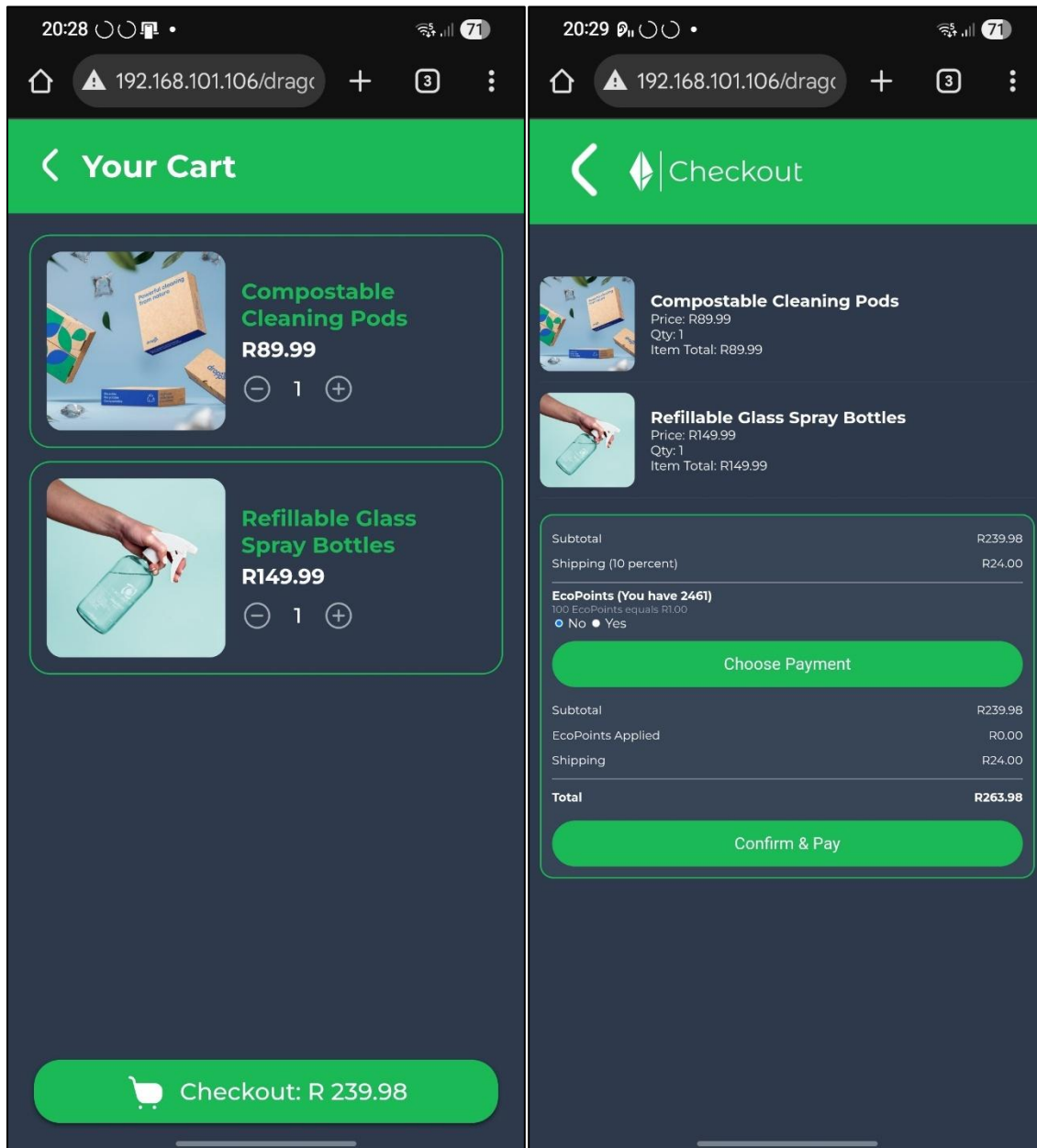
Complete

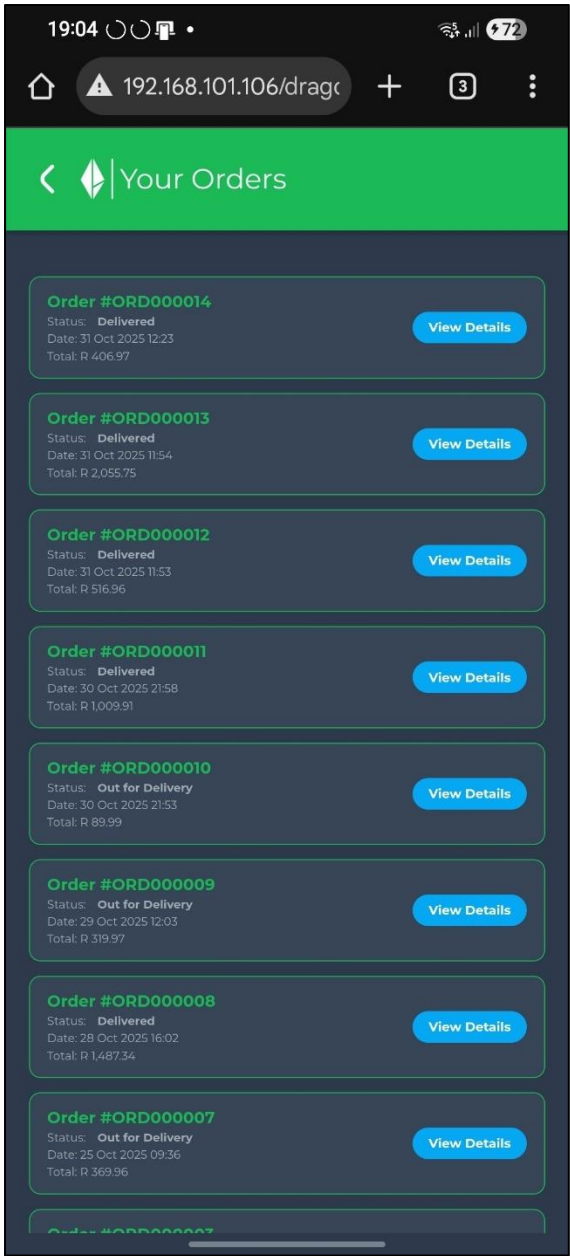
Main Pages





Your Cart, Checkout and Your Orders





Subscription and edit information

19:04 72

192.168.101.106/dragc

Subscription

Biodegradable Trash Bags (20 Pack)

Quantity per subscriber: 1

R99.99

Eco Dish Soap Refill

Quantity per subscriber: 1

R89.99

Eco-Friendly Laundry Detergent (1L)

Quantity per subscriber: 1

R129.99

Natural Deodorant Stick

Quantity per subscriber: 1

R89.99

Organic Hand Wash

Quantity per subscriber: 1

R79.99

Plant-Based Body Soap

Quantity per subscriber: 1

R59.99

Subscribe Now

19:04 72

192.168.101.106/dragc

Edit Information

Jaydyn

Greyling

jaydyn12@outlook.com

824328141

48 Richard avenue, Homestead

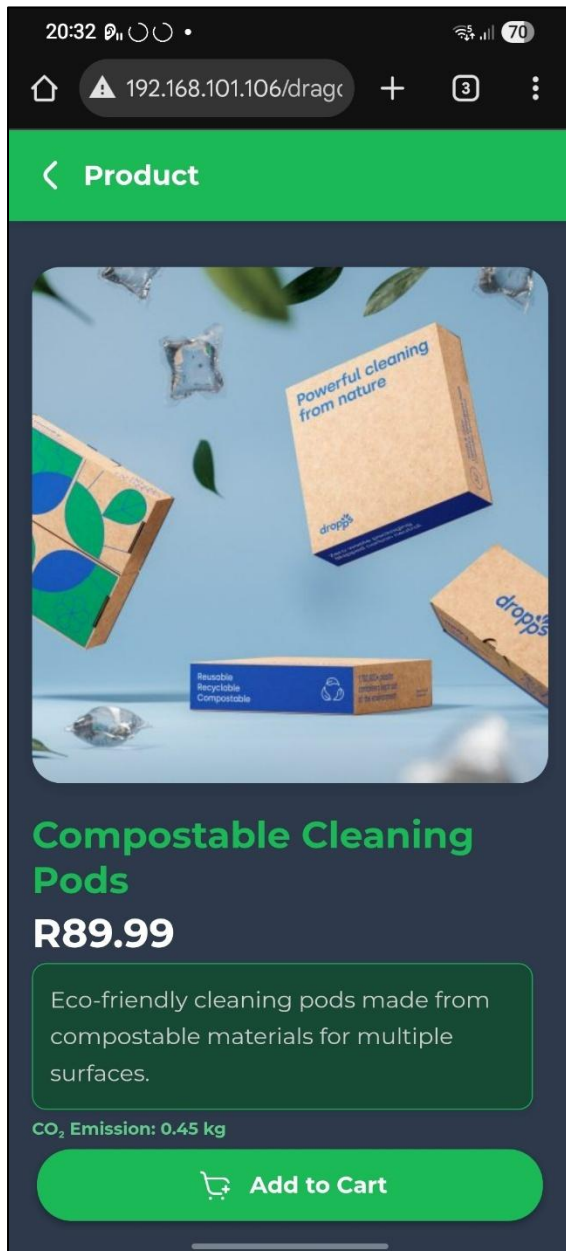
Germiston

JHB

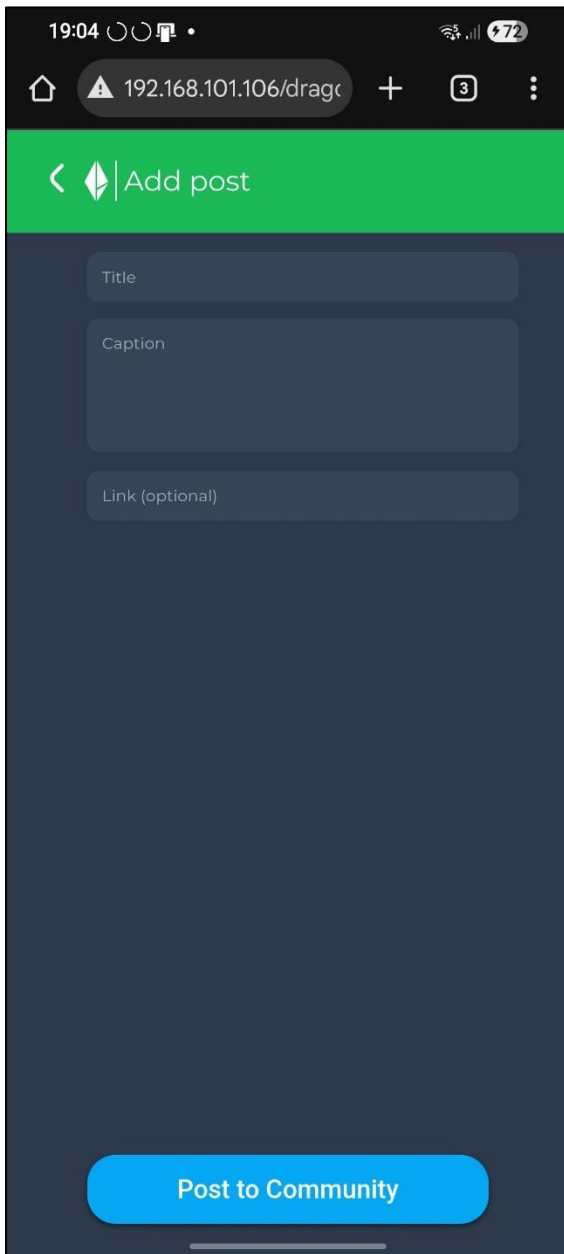
Gauteng

1401

Save Changes



Add post



The image shows a mobile application interface for adding a post. At the top, there is a status bar with the time 19:04, signal strength, and battery level at 72%. Below the status bar is a browser-like address bar showing the URL 192.168.101.106/dragc. The main header is a green bar with a back arrow, a logo, and the text 'Add post'. The form consists of three text input fields: 'Title', 'Caption', and 'Link (optional)'. At the bottom, there is a large blue button labeled 'Post to Community'.

19:04 72

192.168.101.106/dragc

< Add post

Title

Caption

Link (optional)

Post to Community

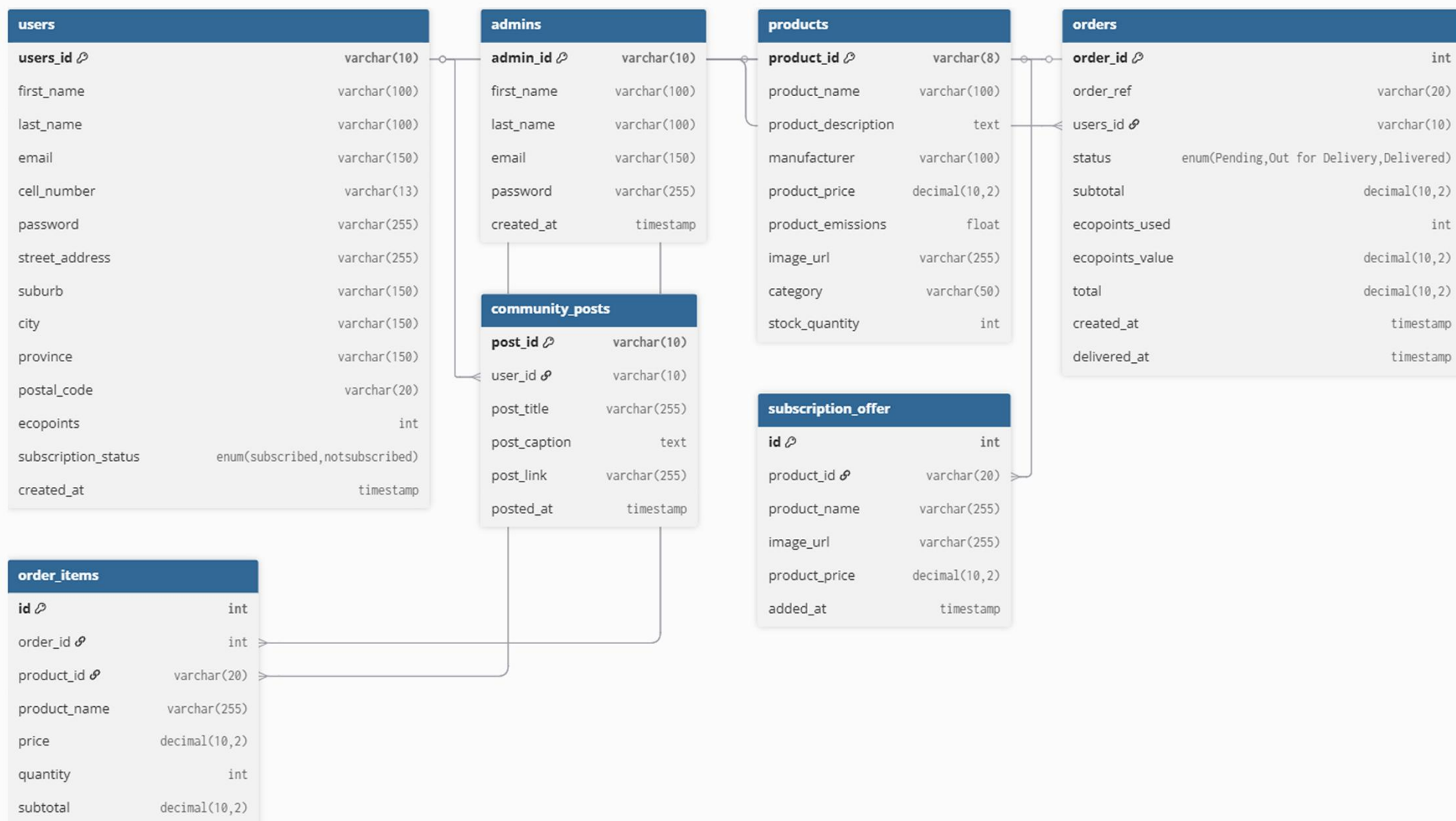
Goals & Objectives

Class Responsibility Collaborator (CRC) Cards

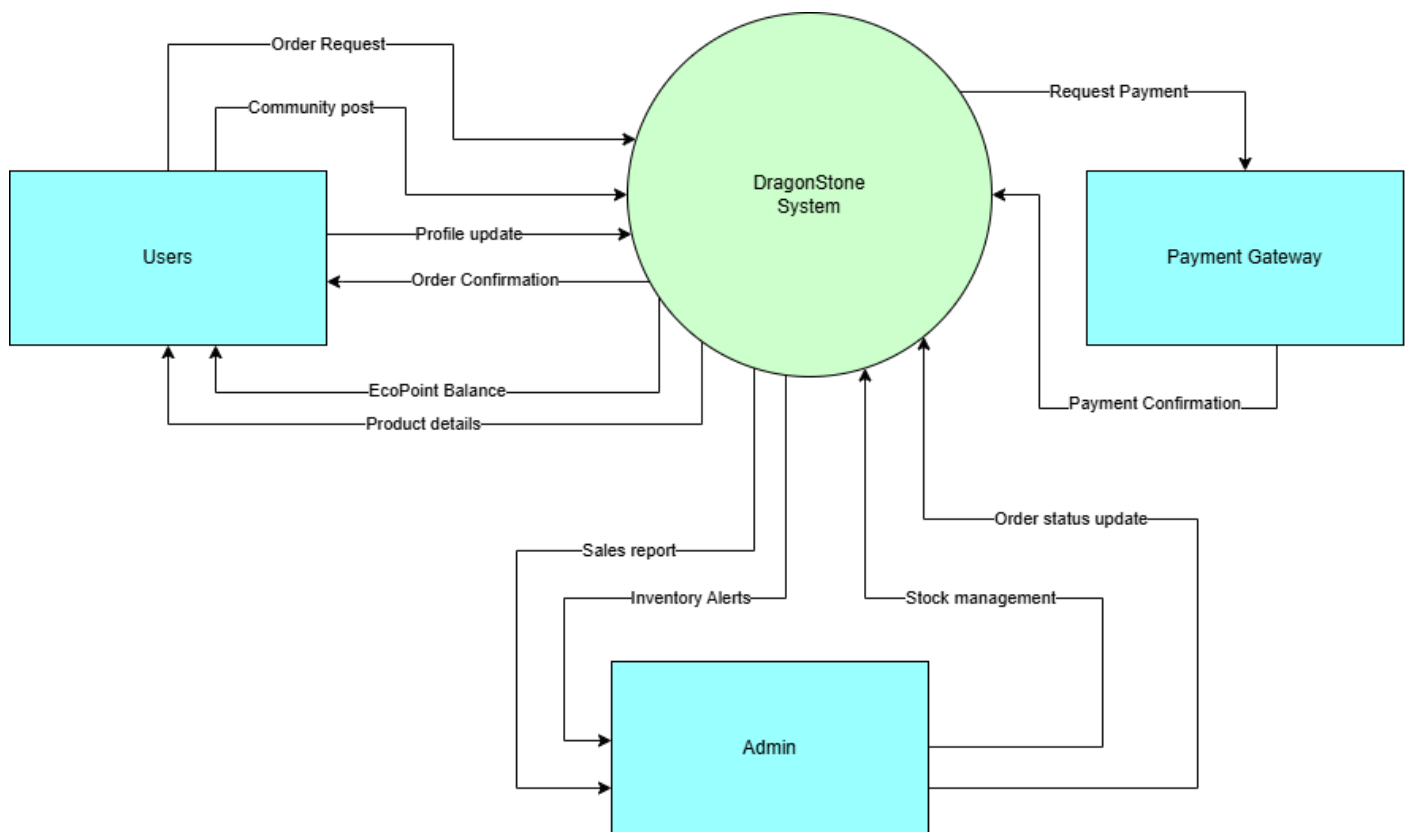
DragonStone - CRC Cards

C User ○ user_id: String ○ first_name: String ○ last_name: String ○ email: String ○ password: String ○ ecopoints: int ○ subscription_status: enum ○ signUp ○ login ○ addPost ○ placeOrder	C Admin ○ admin_id: String ○ first_name: String ○ last_name: String ○ email: String ○ password: String ○ login ○ addProduct ○ updateStock ○ moderatePosts ○ manageOrders	C Product ○ product_id: String ○ name: String ○ description: String ○ price: Decimal ○ emissions: Float ○ category: String ○ stock_quantity: int ○ calculateEmissions ○ updateStock
C Order ○ order_id: Int ○ order_ref: String ○ user_id: String ○ status: Enum ○ subtotal: Decimal ○ total: Decimal ○ createOrder ○ updateStatus	C CommunityPost ○ post_id: String ○ user_id: String ○ title: String ○ caption: Text ○ posted_at: Timestamp ○ addPost	

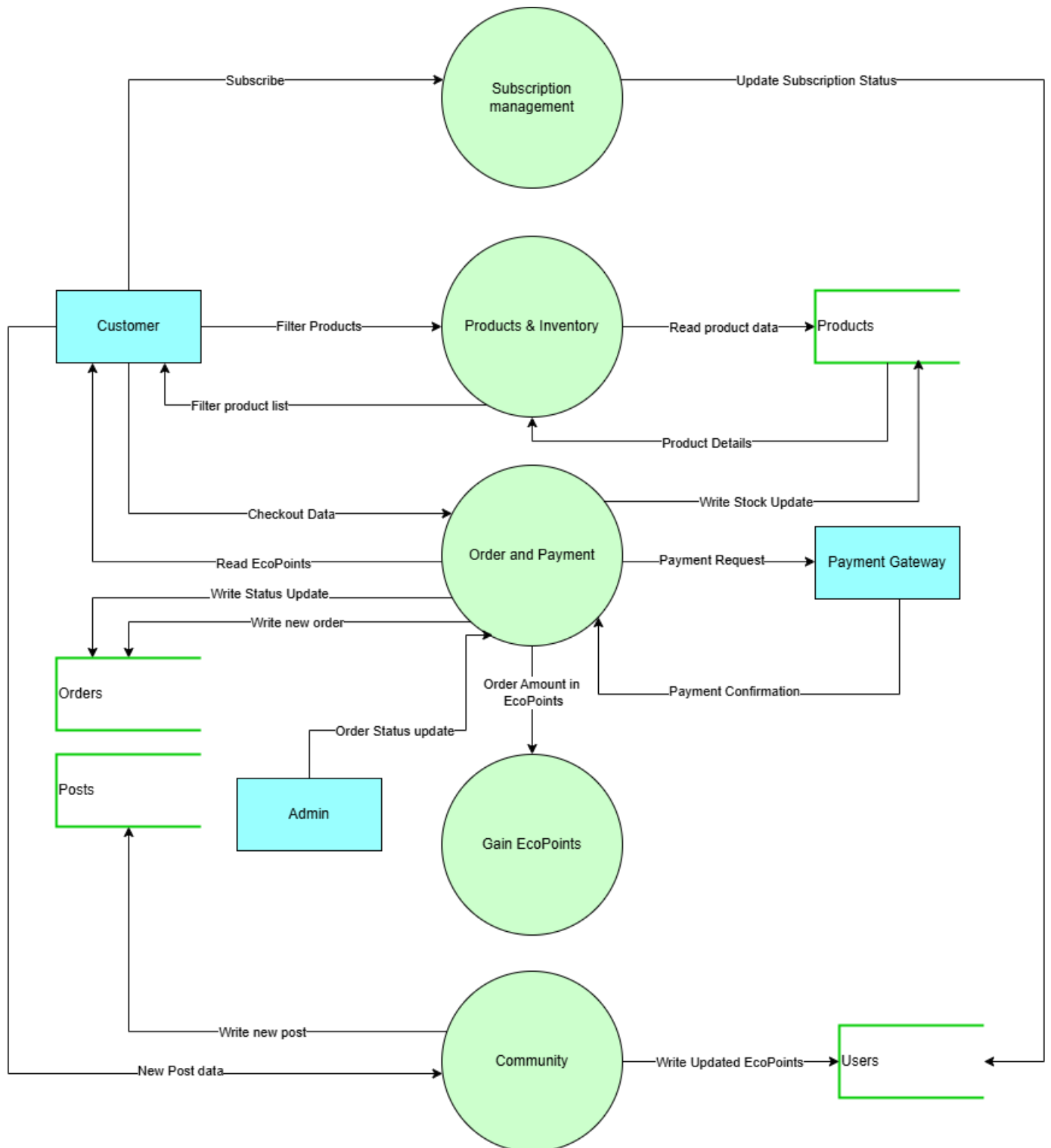
Enhanced Entity Relationship Diagram (EERD)



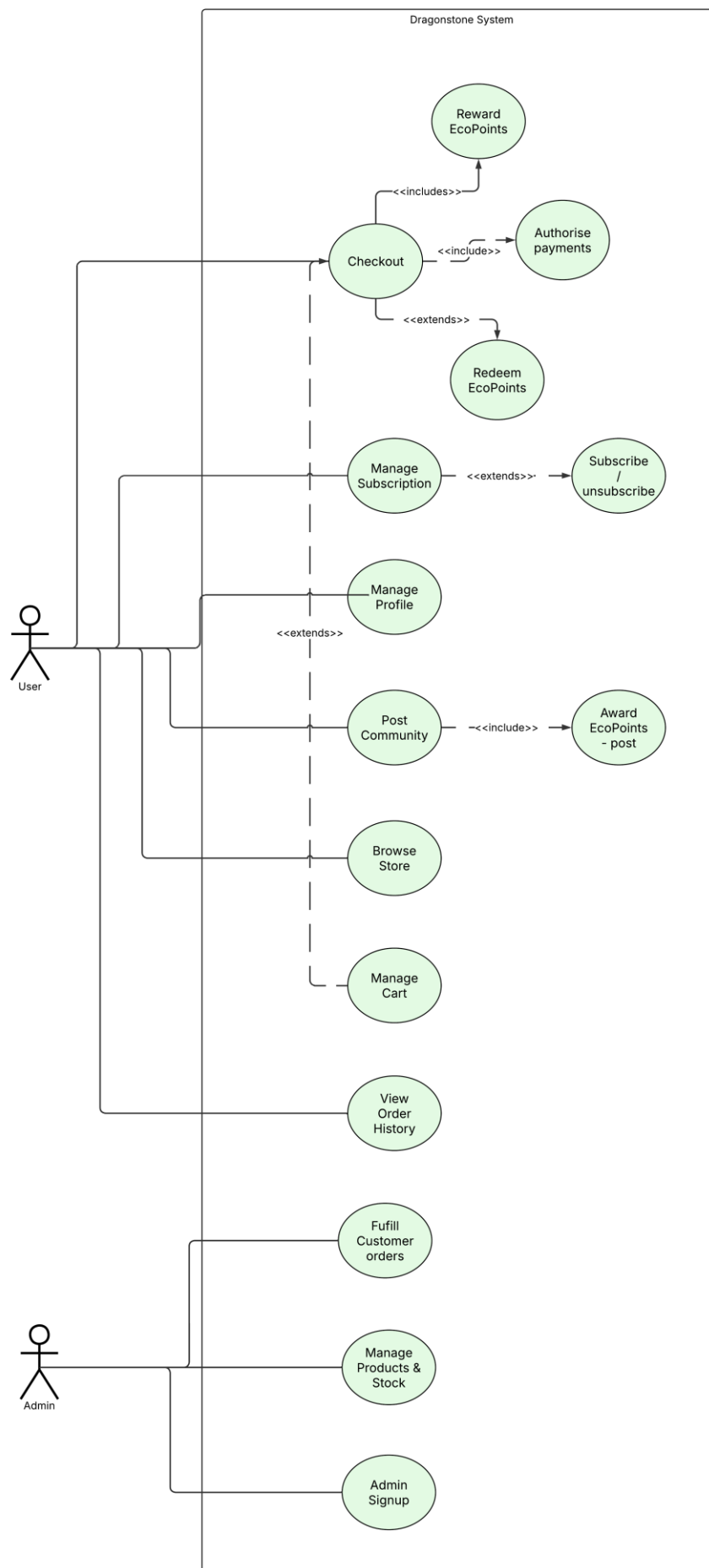
Context Diagram



Data Flow Diagram (DFD)



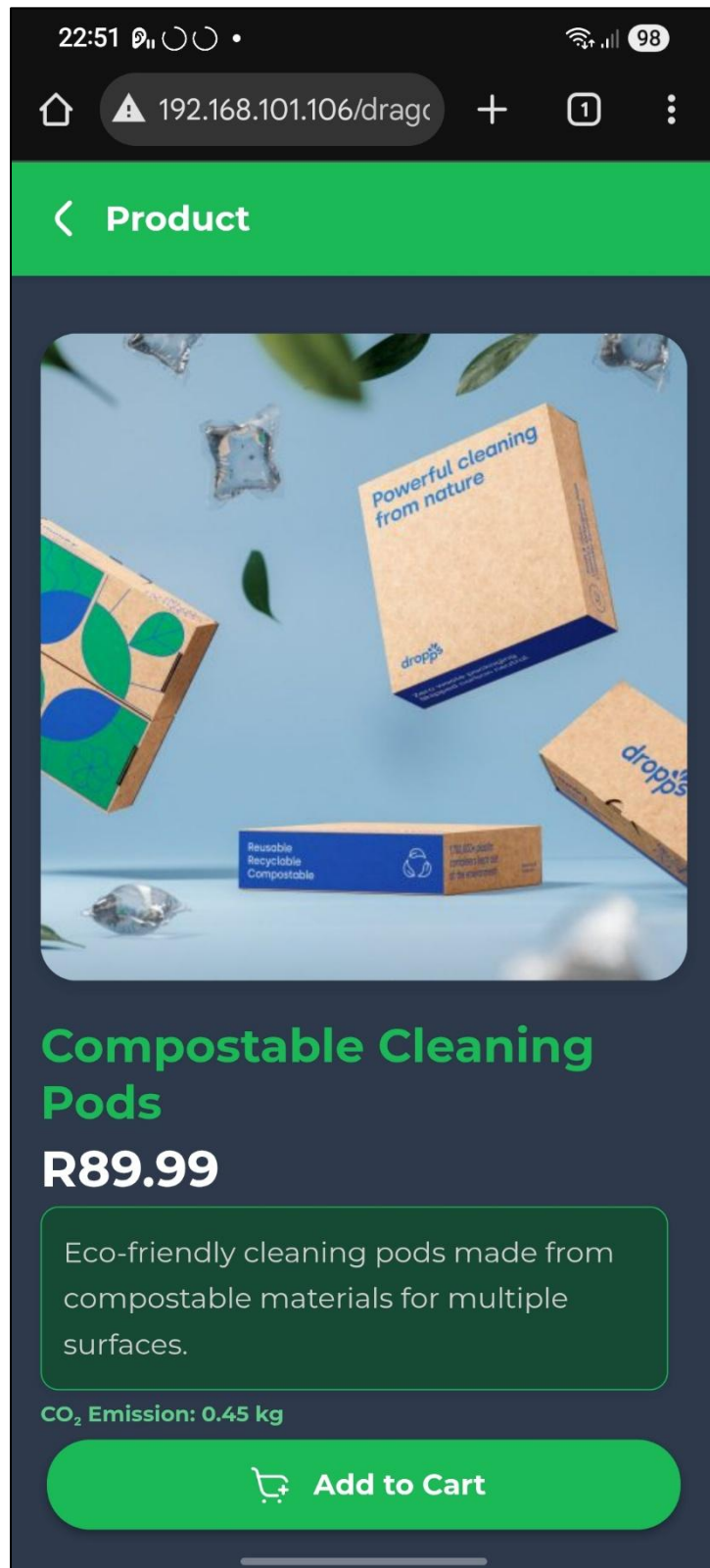
Use Case Diagram



Coding

Sample code has been taken from the product page, **product.php**.

Screenshot



PHP

```
<?php
```

```
include 'db_connect.php';

if (session_status() === PHP_SESSION_NONE) session_start();

if (!isset($_SESSION['cart'])) $_SESSION['cart'] = [];

if (!isset($_GET['id'])) die("Product not found.");

$product_id = $_GET['id'];

$stmt = $conn->prepare("SELECT * FROM products WHERE product_id = ?");
$stmt->bind_param("s", $product_id);
$stmt->execute();
$product = $stmt->get_result()->fetch_assoc();
if (!$product) die("Product not found.");

$inCart = in_array($product_id, $_SESSION['cart']);

?>
```

This PHP code retrieves the product details from the MySQL database using a prepared statement, ensuring security against SQL injection. It checks if the product is already in the user's session-based cart to determine whether to show "Add to cart" or "Remove from cart" on the floating cart button.

HTML

```
<div class="product-section">

  <div class="product-image-container">

    ">

  </div>

  <div class="product-title"><?= htmlspecialchars($product['product_name']); ?></div>
  <div class="product-price">R<?= number_format($product['product_price'], 2); ?></div>
  <div class="product-description"><?= nl2br(htmlspecialchars($product['product_description']));
?></div>

  <div class="product-emission">CO2 Emission: <?= $product['product_emissions']; ?> kg</div>
</div>
```

The HTML structure displays all product details in a clean, readable layout. It includes the image, title, price, detailed description, and CO₂ emissions.

CSS

```
.product-title {  
    font-size: 65px;  
    font-weight: 700;  
    color: #1DB959;  
}  
  
.product-price {  
    font-size: 70px;  
    color: #FFFFFF;  
    font-weight: 700;  
}  
  
.product-description {  
    background-color: #184A35;  
    border: 2px solid #1DB959;  
    color: #D0D0D0;  
    font-size: 40px;  
    border-radius: 25px;  
    padding: 30px;  
}  
  
.floating-cart {  
    background-color: #1DB959;  
    color: #fff;  
    border-radius: 60px;  
    height: 125px;  
    display: flex;  
    align-items: center;  
    justify-content: center;  
}
```

The CSS gives the product page its clean, modern green-and-dark theme. The floating cart button stands out in bright green, while the product section uses a muted blue-grey background to emphasize the product content.

JavaScript

```
function toggleCart(productId) {  
    const button = document.querySelector('.floating-cart span');  
    const icon = document.querySelector('.floating-cart img');  
    const currentAction = button.textContent.includes('Remove') ? 'remove' : 'add';  
  
    fetch('update_cart.php', {  
        method: 'POST',  
        headers: {'Content-Type': 'application/x-www-form-urlencoded'},  
        body: 'product_id=' + productId + '&action=' + currentAction  
    })  
    .then(r => r.json())  
    .then(data => {  
        if (currentAction === 'add') {  
            button.textContent = 'Remove from Cart';  
            icon.src = 'assets/icons/removeCart_icon_alt.png';  
            showToast('Added to Cart');  
        } else {  
            button.textContent = 'Add to Cart';  
            icon.src = 'assets/icons/addCart_icon.png';  
            showToast('Removed from Cart');  
        }  
        sessionStorage.setItem('cartUpdated', 'true');  
    });  
}  
  
function showToast(message) {  
    const toast = document.getElementById('toast');  
    toast.textContent = message;  
    toast.classList.add('show');  
    setTimeout(() => toast.classList.remove('show'), 1800);  
}
```

This JavaScript code handles the real-time cart updates. When a user adds or removes a product, it sends an AJAX request to `update_cart.php` and updates the floating button and icon instantly. The `showToast()` function displays a brief popup message to confirm the action.

MySQL Sample Tables

	Table 	Action	Rows 	Type
☐	admins	  Browse  Structure  Search  Insert  Empty  Drop	1	InnoDB
☐	community_posts	  Browse  Structure  Search  Insert  Empty  Drop	1	InnoDB
☐	orders	  Browse  Structure  Search  Insert  Empty  Drop	9	InnoDB
☐	order_items	  Browse  Structure  Search  Insert  Empty  Drop	28	InnoDB
☐	products	  Browse  Structure  Search  Insert  Empty  Drop	42	InnoDB
☐	subscription_offer	  Browse  Structure  Search  Insert  Empty  Drop	6	InnoDB
☐	users	  Browse  Structure  Search  Insert  Empty  Drop	3	InnoDB
7 tables		Sum	90	InnoDB

Figure 1 MySQL Database tables

 		product_id	product_name	image_url	product_description	manufacturer	product_price	product_emissions	category	stock_quantity
☐	 Edit  Copy  Delete	ITM00001	Compostable Cleaning Pods	assets/images/cleaning_pods.jpg	Eco-friendly cleaning pods made from compostable m...	GreenClean Co.	89.99	0.45	Cleaning & Household Supplies	92
☐	 Edit  Copy  Delete	ITM00002	Refillable Glass Spray Bottles	assets/images/glass_spray_bottle.jpg	Stylish and reusable glass bottles for your cleani...	EcoSpray	149.99	0.35	Cleaning & Household Supplies	997
☐	 Edit  Copy  Delete	ITM00003	Plant-Based Laundry Detergent Sheets	assets/images/laundry_sheets.jpg	Concentrated detergent sheets that dissolve comple...	PureWash	129.99	0.6	Cleaning & Household Supplies	992
☐	 Edit  Copy  Delete	ITM00004	Wool Dryer Balls	assets/images/wool_dryer_balls.jpg	Natural wool balls that reduce drying time and sta...	WoolWorks	99.99	0.2	Cleaning & Household Supplies	992
☐	 Edit  Copy  Delete	ITM00005	Biodegradable Trash Bags	assets/images/biodegradable_bags.jpg	Strong and leak-proof trash bags that decompose na...	GreenBin	79.99	0.5	Cleaning & Household Supplies	1001
☐	 Edit  Copy  Delete	ITM00006	Natural Air Purifying Charcoal Bags	assets/images/charcoal_bags.jpg	Charcoal-based deodorizing bags that absorb moistu...	PureAir	119.99	0.4	Cleaning & Household Supplies	1000

Figure 2 MySQL Product Table

← T →			users_id	first_name	last_name	email	cell_number	password	street_address	suburb	city	province	postal_code	ecopoints	subscription_status	created_at	
☐	 Edit	 Copy	 Delete	USR00001	Jaydyn	Greyling	jaydyng12@outlook.com	824328141	\$2y\$10\$G0gLulj/3Y7c	48 Richard avenue, Homestead	Germiston	JHB	Gauteng	1401	1806	notsubscribed	2025-10-21 13:39:37
☐	 Edit	 Copy	 Delete	USR00002	Gabby	Versfeld	gabby@mail.com	0123456789	\$2y\$10\$JulxcBYQ5i.bl	44 Alsatian Rd, Glen Austin AH	Midrand	Midrand	Gauteng	1685	669	notsubscribed	2025-10-21 17:54:37
☐	 Edit	 Copy	 Delete	USR00004	Heidi	Greyling	heidi@basf.com	27795051334	\$2y\$10\$BSpBa1fH6c4	Cnr Nywerheid Street	Elandsfontein	Germiston	Gauteng	1406	428	notsubscribed	2025-10-21 20:31:52

Figure 3 MySQL Users Tabl

Table of Figures

Figure 1 Login	Error! Bookmark not defined.
Figure 2 Sign up (Personal info)	Error! Bookmark not defined.
Figure 3 Sign up (Address info)	Error! Bookmark not defined.
Figure 4 Shop	Error! Bookmark not defined.
Figure 5 Community.....	Error! Bookmark not defined.
Figure 6 Profile.....	Error! Bookmark not defined.
Figure 7 Your Cart.....	Error! Bookmark not defined.
Figure 8 Checkout.....	Error! Bookmark not defined.
Figure 9 Order Success	Error! Bookmark not defined.
Figure 10 Your Orders.....	Error! Bookmark not defined.
Figure 1 Product page (Sample code)	Error! Bookmark not defined.
Figure 2 MySQL Database tables	24
Figure 3 MySQL Product Table	24
Figure 4 MySQL Users Table	25